

Distributed Communication: Definition and Planning (Partial Delivery 1)

Instructions

In this assignment, you will deepen your knowledge of distributed communication technology, paying special attention to both the theory and architecture of this fundamental technology for modern systems.

Activities:

1. **In-depth research:** Conduct a detailed investigation into how distributed communication technology works. Carefully analyze the protocols used to enable communication between systems, understanding how message exchange occurs and how the headers of these messages are structured. Seek to understand the importance of these elements to ensure that different systems can communicate efficiently, securely, and reliably. Explain how these protocols and structures contribute to interoperability and the integrity of transmitted data.
2. **AI scenario definition:** Clearly and precisely describe the Artificial Intelligence scenario that will be used in the work. Specify the types of data that will be transmitted in this context, such as JSON files, tensors (essential data structures in AI), or text streams. Briefly justify the relevance of this type of data to the chosen scenario, showing the relationship between the nature of the data and the needs of the AI system.
3. **Communication diagram design:** Create an illustrative diagram representing the communication flow in the defined scenario. Choose between the Client/Server model or the Publisher/Subscriber (Pub/Sub) model, according to your work's context. Clearly demonstrate how data circulates between system participants, indicating the role of each component and highlighting the interactions between them. Ensure the diagram is comprehensible and accurately reflects the proposed architecture.

Delivery:

Assemble a PDF document containing:

- The names of the group members
- The chosen distributed communication technology
- References used in the research (books, articles, websites, etc.)
- A summarized description of the AI scenario (POT), explaining its functioning and the purpose of the communication.

Put effort into the explanation and organization of the document. Detail each part of the work, demonstrating mastery of theoretical concepts and the architecture of the studied technology. Ensure all information is well-founded and presented clearly, evidencing your understanding of the topic.

Distributed Communication: Development and Implementation (Partial Delivery 2) - NEED TO DO THIS PART

Instructions

Focus: Implementation (Coding).

Activities:

- Programming environment setup (Python, Node.js, etc.).
- Use of AIs (Gemini/ChatGPT/Copilot) for boilerplate generation and code debugging—if you prefer, I can help.
- Implementation of the "Core" of communication (e.g., Create Kafka Broker or define .proto file in gRPC).
- Connectivity tests between remote machines/processes.

Delivery: Preliminary .pptx file with the architecture and screenshots of the code/initial functioning.

Distributed Communication: Seminar and Final Demonstration

Due May 5, 2026 11:59 PM

Instructions

Focus: Knowledge dissemination.

Activities:

- Seminar presentation (maximum 15-20 min per group).
- Live demonstration of the POT or via recorded video.
- Discussion on the advantages and limitations of the technology for large-scale systems.

Delivery: GitHub repository link with the complete code, detailed README, and final presentation version.

